

The Resonance Chain

From the Crystal's Ring to Atoms: the Golden Mass Ladder, its
Placements,
the Deviation Decomposition, and the Fenced Open Problem of Rung
Selection

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Status: Preprint (not peer-reviewed). This paper is the citable record underpinning the programme's Standard-Model ledger simulator (`scripts/sm_ledger.py`) and the public web explorer; every physics number either of those displays is derived or recorded here or in the paper cited at its point of display (the cosmological-constant entry cites the cosmology branch). Computational claims: `scripts/explore_shell_rule.py` (11/11), `scripts/explore_geometric_ladder.py` (3/3), `scripts/explore_offset_rationals.py` (5/5), with upstream inputs from the suites of Papers LIII–LVIII. Grades used throughout: DERIVED / CONDITIONAL / PLACED / OPEN, in the sense of §1.

Abstract

This paper assembles, in one citable place, the framework's complete current account of particle masses and the route from the substrate's vibrations to atoms — including, with equal care, what is derived, what is placed, and what is open. The account is a five-link chain. (1) The crystal's resonant spectrum at fixed grain is exact and closed-form (Paper LV). (2) Mass is closure frequency: a particle is a self-closing mode whose rest frequency is set by the coherence length at which it closes, an identification verified dynamically (Paper LIII). (3) Given the φ -self-similar refinement of the substrate (a named premise of the cosmogenesis layer) and the verified closure-frequency identification, the mass scale is geometric: $m(N) = m_P \varphi^{-N}$, one golden step per refinement level; anchored on the single structural rung $N_\mu = 96 = 24 \times 4$ (Hypothesis H-shell-96, Paper LII), every rung value is parameter-free, and the thirteen Standard-Model placements land within half a rung across nineteen orders of magnitude — the muon at parts-per-million, the Z at -2.3% , the worst placements near $\pm 25\%$. (4) *Which* rung a particle takes is the open link, and we fence it at proof grade: the separability obstruction refutes every additive *and every affine* rule built from static quantum numbers and a universal generation profile (curvatures $(5, 3, -2)$ and anchored ratios $(1.545, 1.769, 2.333)$ both type-distinct); per-type linear rules fail; an information budget (≤ 25 parameter bits for the nine charged-fermion integers) sets the bar any future rule must clear; eight geometry- and prime-derived level sieves test null against exact hypergeometric null models after look-elsewhere correction, with the surviving protocol target recorded (GL3:

a derived sieve capturing all nine shells at span-density $\leq 1/3$ would clear the pre-registered bar); a one-loop-log dressing form is excluded within the lepton sector; and a sector-limited offset–structure correlation that failed its out-of-sample test is recorded as a lead, not a law. The deviations themselves decompose into four recorded causes: uniformly finer rungs are disfavoured (sub-rung positions are chance-like against half- and quarter-shell grids; non-uniform structures fall under the sieve protocol); the W’s miss is the Z’s placement plus the weak mixing angle exactly ($m_W = m_Z \cos \theta_W$, so the electroweak sector carries one placement, one already-named open, and the Higgs); the light-quark and proton comparisons are scheme and binding artifacts (MS-bar conventions; $\sim 99\%$ QCD binding energy) rather than failures of a bare spectrum; and the clean remainder — smallest for the cleanest particles (mean |offset|: leptons $0.071 < \text{EW } 0.184 < \text{quarks } 0.288 < \text{composite } 0.462$) — has the profile of interaction dressing, blocked behind the underived coupling sector. (5) The nonrelativistic Coulomb atom then follows as a theorem of the already-derived layer: with the framework’s own electron and conditional α , hydrogen’s ground state computes to -14.14 eV against the -13.61 reference, the discrepancy being, to leading order, the electron’s $+3.9\%$ mass-input error; with the measured electron the Coulomb-level values return at the model’s own accuracy. Nothing in this paper is fitted; every negative result is published as a constraint.

1 Position, and the grading convention

Papers LIII–LVIII derive the framework’s kinematic and dynamical structure: the gravity chain, the gauge inventory, chirality, and the charge lattice. What none of them assembles is the mass story — the part a reader of the programme’s ledger simulator most needs: where the ladder comes from, how good the placements actually are, why most deviations are large, and what stands between the present state and derived masses. This paper is that record. It consolidates, with proofs and machine-checked nulls, material first developed in the repository documents `resonance-mechanism.md`, `shell-rule-wo.md` (work order WO-SHELL-OFFSET-001), and `sm-closeability-audit.md`.

Grades, used on every claim: DERIVED (proved or machine-verified from the framework’s structure); CONDITIONAL (derived under one named open hypothesis); PLACED (a slot read from data and recorded as postdiction); OPEN (not derived, no mechanism in hand). The discipline behind the grades is the programme’s standing rule: golden-ratio expressions are flexible enough to fit any spectrum, so nothing below is fitted, criteria are fixed before judging, look-elsewhere is accounted, and negatives are published.

2 The chain: links 1–3 (links 1–2 derived; link 3 conditional)

Link 1 — the ring. The substrate’s resonant spectrum at fixed grain is closed-form and exact: $\lambda_k = 12 \left(1 - \frac{\sin((k+1)\pi/5)}{(k+1)\sin(\pi/5)}\right)$ with multiplicities $(k+1)^2$, $k = 0..5$ — the harmonics of a 3-sphere (Paper LV [Smart, 2026c]; machine precision). These give fields their shapes; they are not yet masses.

Link 2 — mass is closure frequency. In the framework’s reading, a particle is a localized mode that closes on itself and its rest frequency is its mass. The identification is verified for the substrate’s massive modes: the massive end of the response family $C_\varphi = L + \varphi^{-2}I$ yields modes obeying $(\square - \mu^2)F = 0$ with $\mu = \varphi^{-1}$ — rest frequency set

by the closure coherence length — with the normalization $\omega_0 = c\mu = c/\ell$ fixed by the dispersion of the substrate wave operator; such a mode’s envelope obeys the Schrödinger equation with exactly that mass, and its inertia equals its gravitating energy on one simulated wavepacket (Paper LIII [Smart, 2026b], items T11–T12). That *each Standard-Model particle* is such a mode is the framework’s reading, carried as interpretation, not separately proven per particle.

Link 3 — the golden ladder (conditional on the refinement premise).

Proposition 1 (The ladder, from two named premises). *Assume: (P1) the substrate’s refinement tower rescales its geometry by one factor of φ per level — the φ -recursive multi-shell refinement stated as Theorem 5.4 of the cosmogenesis layer and used throughout the rung ladder; this premise is stated in those documents with the full recursion deferred (it is itself construction-grade there, not theorem-grade), and is imported here as a named premise; (P2) a mode’s rest frequency is set by its closure coherence length — the identification verified dynamically in Paper LIII (items T11–T12). Then a mode closing at depth N closes at coherence length $\propto \varphi^N$ and rest mass*

$$m(N) = m_P \varphi^{-N}.$$

The geometric spacing of the ladder is thereby CONDITIONAL on (P1) — with (P2) carrying machine verification — and the spacing is the only part of the mass story this proposition claims; the integer N for a given particle is not derived (§4).

Hypothesis 2 (H-shell-96, the anchor). The muon’s shell is exactly the structural integer $96 = 24 \times 4$. Provenance: adopted by the muon $g-2$ analysis (Paper LII [Smart, 2026a]) prior to and independently of the mass-ladder questions; the measured offset is -2×10^{-4} shells (parts-per-million in mass). Under this hypothesis $m_P = m_\mu \varphi^{96} = 1.2210 \times 10^{19}$ GeV, and Newton’s constant acquires its conditional anchor — with Paper LVII’s [Smart, 2026e] caveats imported in full: the integer was first read off using measured G (the non-circularity rests on Paper LII’s prior usage), and the look-elsewhere-corrected chance probability is ≈ 0.006 under a stated reference model.

3 The placements (placed; postdiction disclosed)

With the anchor fixed, every particle’s rung value is a parameter-free number. The thirteen placements (PDG 2024 central values; shells $N = \text{round}(\log_\varphi(m_P/m))$, first recorded in the cascade mass analysis):

particle	N	$m_P \varphi^{-N}$ [GeV]	PDG [GeV]	dev.	offset [shells]
electron	107	5.309×10^{-4}	5.110×10^{-4}	+3.9%	+0.079
muon	96	0.10566	0.10566	(anchor)	-0.0002
tau	90	1.896	1.777	+6.7%	+0.135
up	104	2.249×10^{-3}	2.16×10^{-3}	+4.1%	+0.084
down	102	5.888×10^{-3}	4.67×10^{-3}	+26.1%	+0.481
strange	96	0.10566	0.0934	+13.1%	+0.256
charm	91	1.172	1.273	-8.0%	-0.172
bottom	88	4.964	4.18	+18.7%	+0.357
top	81	144.1	172.8	-16.6%	-0.377
proton	91	1.172	0.9383	+24.9%	+0.462
W	82	89.07	80.38	+10.8%	+0.213
Z	82	89.07	91.19	-2.3%	-0.049
Higgs	81	144.1	125.3	+15.1%	+0.291

Conventions and disclosure: the offsets in the table follow the provenance table of the cascade mass analysis (computed with the measured Planck mass), which is why the muon row shows -0.0002 ; under the anchored convention $m_P = m_\mu \varphi^{96}$ the muon offset is identically zero and all other entries shift by that same 2×10^{-4} . Except the muon’s, the integers were read from data — these are PLACED, postdictions, and the programme’s own evidence ledger has always marked the mass placements as noticed-after-the-fact. The calibration that makes the table meaningful: rungs are a factor φ apart, so half a shell (+27.2% upward or -21.4% downward in mass) is the maximum possible miss given correct integers — and the placements hold within that across nineteen orders of magnitude, with the muon and Z sharp. Neutrino masses are not placed at all: OPEN.

Remark 3 (Relation to Paper V’s 13-particle correspondence model). Paper V [Smart, 2026g] reports the same thirteen particles at 0.014% average relative error (maximum 0.047%) — three orders of magnitude tighter than this paper’s bare-ladder placements. The two results are not in tension; they sit at opposite ends of a precision-versus-parameters trade-off, and Paper V’s own hardened text records its cost: a non-unique rule-based (S, w) assignment scheme, three structural anchors, ten chain-ratio correspondences of which *five are reverse-identified post hoc*, and two model-level conventions, with the verdict “high-accuracy numerical correspondences ... predictive content limited by the assignment-scheme freedom and by the post hoc identification of the chain ratios,” and with the tightly-measured entries exceeding PDG uncertainty by orders of magnitude (within-framework correspondence, not within-uncertainty match). In the present paper’s language, Paper V’s rational chain-ratio layer plays a role *analogous* to the offset layer that §4 names open — each chain ratio acts as one particle’s discrete dressing factor, found by matching rather than derived — though the two constructions are not identical: Paper V’s exponents are not these shell integers, its bookkeeping (anchor $\times \varphi^{\text{int}} \times \text{ratio}$) differs, and its particle set (including the neutron) is not the same as this table’s. The bare ladder refuses to fill that layer post hoc and prices the refusal honestly (offsets up to half a shell); Paper V fills it and prices the filling honestly (assignment freedom, post hoc ratios). The open problem is of the same kind in both descriptions: derive the discrete dressing layer — Paper V’s ratios there, these offsets here — from the coupling sector. A successful derivation would have to reproduce the measured masses and would thereby explain or supersede Paper V’s ratio table.

4 Link 4, fenced: the rung-selection problem (open)

The integer-selection rule is the open link, and the work order WO-SHELL-OFFSET-001 fenced it on every accessible side. Each result below is machine-checked (items SR1–SR11, GL1–GL3).

Theorem 4 (Separability obstruction; SR1, SR11). *Write the charged-fermion shells by type $T \in \{L, U, D\}$ and generation $g \in \{1, 2, 3\}$. Two independent invariants refute two nested rule classes, for any universal generation function f and coefficients depending on arbitrary static quantum numbers:*

- (a) (additive) *In $N = a_T + f(g)$ the generation curvature $N_T(3) - 2N_T(2) + N_T(1) = f(3) - 2f(2) + f(1)$ is type-independent; the observed curvatures are $(5, 3, -2)$ — distinct. Refuted.*
- (b) (affine, strictly stronger) *In $N = a_T + b_T f(g)$ the anchored ratio $(N_T(3) - N_T(1)) / (N_T(2) - N_T(1)) = (f(3) - f(1)) / (f(2) - f(1))$ is type-independent; the observed ratios are $(1.545, 1.769, 2.333)$ — distinct. Refuted, subsuming (a) (the case $b_T \equiv 1$) and the multiplicative case ($a_T \equiv 0$).*

What this leaves open, within the static class, are rules with genuine type-generation coupling beyond affine; such rules need at least the per-type-quadratic parameter count and are then governed by the protocol budget of Proposition 5.

Proposition 5 (Cheapest exact class is inadmissible; SR2–SR3). *Within-type generation steps are unequal for all three types ($L : -11, -6$; $U : -13, -10$; $D : -6, -8$), so per-type linear-in- g rules (six parameters) admit no exact fit; within the per-type polynomial hierarchy the cheapest exact class is quadratic-per-type — nine parameters for nine integers, zero compression. Information budget (a pre-registered protocol choice, not a theorem): the nine integers span 27 values (≈ 43 bits); the programme commits to admitting only rules spending ≤ 25 parameter bits (compression $\geq 1.7\times$). Any future proposal below this bar is rejected by protocol, whatever its fit.*

Proposition 6 (Sieve scan; GL1–GL3). *A sieve — geometry generating a set of allowed levels, particles occupying allowed levels — is not a function of quantum numbers, so Theorem 4 does not exclude it; it is the surviving class among those tested. Eight geometry- and prime-derived sieves (multiples of 8 and of 6; $24a + 5b$ cosets; the H_4 -degree ladder; $96 \pm E_8$ exponents; primes and prime powers; split-prime resonances $p \equiv \pm 1 \pmod{5}$; pentagonal classes) were scored against exact hypergeometric null models with Bonferroni correction and a density bound: **all null** (best: multiples of 8, corrected $p = 0.23$). The surviving quantitative door is recorded: a derived level set capturing all nine observed shells at span-density $\leq 9/27$ clears the bar decisively — derive first, blind to the table; score once.*

Proposition 7 (Exclusions and a recorded lead; SR7–SR8). *The one-parameter dressing form $\delta = c Q^2 \ln(m_P/m)$ is monotone in the shell; the lepton offsets $(+0.079, -0.0002, +0.135)$ are not: that specific form is excluded within the lepton sector (general loop dressings with more structure are not). Separately, signed offsets regress on (Q^2, colour, N) at $R^2 = 0.86$ in-sample (permutation $p = 0.011$; leave-one-out 0.46) but the fermion-fitted law fails catastrophically out-of-sample on the bosons it never saw (Z predicted $+1.34$ vs observed -0.049 , outside the admissible offset range entirely): recorded as a sector-limited lead, not announced as a law.*

The conclusion the fence forces: the integers do not factor through static structure in any additively or affinely separable way — consistent with (though not proof of) their encoding *interacting* dynamics — the closure condition of a self-interacting mode. The dependency order for derived masses is therefore: couplings $\rightarrow$ closure selection $\rightarrow$ shell integers $\rightarrow$ offsets.

5 The deviation decomposition (recorded attributions: null tests, bookkeeping, observation)

The placements' deviations are not one mystery; they decompose into four *recorded attributions*. The evidence types differ and are labelled: a Monte-Carlo null (Proposition 8), an algebraic identity (Proposition 9), and an $n = 13$ observation (Observation 10); none is a derivation of the offsets themselves.

Proposition 8 (Not finer uniform rungs; SR9). *Against uniform half-shell and quarter-shell grids, the thirteen sub-rung positions are statistically indistinguishable from uniform (Monte-Carlo $p = 0.32$ and 0.40 ; integer grid $p = 0.28$). The missing ingredient is not a uniformly finer ladder of the same kind; non-uniform intermediate structures are not excluded by this test (they fall under the sieve protocol of Proposition 6).*

Proposition 9 (The W is the Z plus the mixing; SR10). *The W–Z shell split is $\log_\varphi(m_Z/m_W) = 0.262$, identically $\log_\varphi(1/\cos\theta_W)$ by the on-shell (tree-level) electroweak relation $m_W = m_Z \cos\theta_W$ (equivalently $\sin^2\theta_W = 0.2231$ from these masses, versus the PDG on-shell 0.2233). This is bookkeeping, not discovery — but it is correct bookkeeping: the W carries no independent placement information, and the electroweak sector reduces to one placement (Z, -2.3%), one already-named open (the mixing angle / independent hypercharge, Papers LVI [Smart, 2026d] and LVIII), and the Higgs.*

Observation 10 (Scheme and binding artifacts; SR6). Mean $|\text{offset}|$ by sector is monotone: leptons $0.071 < \text{EW bosons } 0.184 < \text{quarks } 0.288 < \text{composite } 0.462$. The ladder is sharpest where mass is a clean pole observable and worst where the comparison number is not a physical mass at all — an ordering *consistent with* (not demonstrating) the dressing reading: light-quark entries are MS-bar scheme conventions, and the proton is $\sim 99\%$ QCD binding energy — quantities no bare spectrum should reproduce. Recorded implication (falsifiable): reformulating the table against pole/physical masses only should sharpen it materially. ($n = 13$; an observation, not a theorem.)

What remains after Propositions 8–9 and Observation 10 is the genuine residual: the lepton-sector offsets (≤ 0.135 shells) and residual quark structure, with the profile of interaction self-energy on a bare spectrum — the framework's analogue of radiative corrections, OPEN, blocked behind the coupling sector. Each particle in the ledger and the public explorer carries its attributed cause (ANCHOR / PLACEMENT / MIXING / SCHEME / BINDING / DRESSING).

Phase D: the primitive-rational offset correspondence (recorded lead, control passed)

Re-reading Paper V's ratio layer through the ladder suggests a concrete candidate shape for the dressing: the correction factor $r = m_{\text{rung}}/m_{\text{meas}}$ of the form $1 \pm a/b$ with b a

named substrate count. The test, specified in full: *alphabet* = $\{1 \pm a/b\}$ with $a \in \{1, 2, 3\}$ and $b \in \{5, 6, 8, 9, 12, 14, 15, 16, 20, 24, 30, 42\}$ (respectively: the BFS shell count, $1/R_I$, the observer rung, an integer Laplacian eigenvalue, the vertex degree, an integer eigenvalue, $R_D(4)$, the info rung, the second-shell population, the 24-cell, the third-shell population/ H_4 Coxeter number, and V_3), pruned to the interval $(0.75, 1.35)$, giving 42 members; *data* = the twelve non-anchor particles of the placements table, with r computed in the anchored convention ($m_P = m_\mu \varphi^{96}$; the muon is excluded as the anchor); *statistic* = the sum of log relative distances to the nearest alphabet member; *null* = the same statistic for offsets drawn uniformly on $(-\frac{1}{2}, \frac{1}{2})$, 2×10^4 Monte-Carlo draws, same alphabet (so alphabet density is controlled by construction). Results (suite `explore_offset_rationals.py`, 5/5): the registered alphabet gives $p = 0.0019$ with 8/12 particles within 0.2% of a member (chance expectation 3.9); a *control* alphabet built the same way from twelve non-primitive prime denominators — larger, 68 members — comes out null ($p = 0.42$, 5/12): an outcome consistent with the offsets preferring the primitive denominators, though the single passed control does not prove denominator specificity; the sub-alphabet restricted to Paper-V-native counts $\{5, 6, 12, 15, 20, 24, 30, 42\}$ retains $p = 0.015$; restricting numerators to $a \leq 2$ weakens the signal to $p = 0.067$ (the $a = 3$ forms 19/16 and 23/20 matter; disclosed). The corrected ladder reaches mean $|\text{error}| = 0.198\%$ at ≈ 5.4 bits per particle — bottom 19/16 at -0.001% , tau 16/15, Higgs 23/20, Z 41/42, proton 5/4, top 5/6 = $1 - R_I$ (a form that also appears among Paper V’s forward-traceable ratios, there for the charm).

Status, stated at full strength: *recorded correspondence*, not a derivation, and the quoted p -values are *conditional, post-sight figures*: the alphabet was assembled with partial sight of the data (candidate fractions were noticed for a few particles before the alphabet was frozen), so they quantify closeness under the stated protocol rather than a blind discovery probability. The mitigations are the density-controlled null, the passed control, and the surviving uncontaminated sub-alphabet; the alphabet is hereby **frozen** for any future re-test, and the precision entries are not within experimental uncertainty, so sharper theory — not sharper data — is what can confirm or kill the lead. What it changes: the open offset dynamics acquires a concrete target shape — obtain a/b per particle from the closure mode’s population structure — and Remark 3’s analogy between Paper V’s ratio layer and the offset layer acquires quantitative support.

6 Link 5: atoms as a theorem (derived chain)

Once a mode has a mass, a charge, the Schrödinger equation, and the Coulomb law, the *nonrelativistic Coulomb model of the atom* follows by ordinary quantum mechanics — and each ingredient is in the derived layer: the Schrödinger reduction and Maxwell uniqueness (Paper LIII), the charge lattice and chirality (Paper LVIII [Smart, 2026f]). Quantitatively, with the framework’s conditional $\alpha^{-1} = 137 + \pi/87$ and its *own* electron ($N = 107$, PLACED):

$$E_1 = -\frac{1}{2}\alpha^2 m_e c^2 = -14.14 \text{ eV} \quad (\text{infinite-nuclear-mass Rydberg reference } -13.6057), \quad a_0 = 5.09 \times 10^{-11} \text{ m}$$

and the discrepancy is, to leading order, the electron’s +3.9% mass-input error propagating linearly — substituting the measured m_e returns the infinite-nuclear-mass values $E_1 = -13.6057 \text{ eV}$ and $a_0 = 5.292 \times 10^{-11} \text{ m}$. The standard corrections beyond that model belong to the same derived equations applied at higher order, and to demonstrate

that this is substantive rather than a deferral we compute the first one: replacing m_e by the reduced mass $\mu = m_e/(1+m_e/m_p)$ — the two-body solution of the *same* Schrödinger–Coulomb problem — moves the measured-electron prediction from $E_1 = -13.6057$ eV to $E_1 = -13.5983$ eV against hydrogen’s measured ionization energy 13.5984 eV: the residual falls from 7.3 to 0.13 meV — a factor ≈ 55 — with no new input. Fine structure (order α^2), Lamb and hyperfine effects sit further down the same tower and are not computed here. The claim is scoped accordingly: hydrogen’s Coulomb-level structure is a theorem of the derived layer given the mass input; chemistry downstream needs no further framework input beyond the open mass sector.

7 The simulator as reproducibility artifact

Two executable artifacts mirror this paper exactly and contain no quantity not derived or recorded here: the command-line ledger `scripts/sm_ledger.py` (grades, mass table, charge/coupling/atoms sectors) and the public web explorer (single-file, computing everything in-browser from φ , the anchor, and the derived equations; the ladder drawn at *true* offsets; the decomposition section required; the Link-4 fence displayed). The maintenance rule is part of the record: the simulators are downstream of the verified repository, never ahead of it — a quantity enters them only after it carries a verification item. Checks: `explore_shell_rule.py` (10/10), `explore_geometric_ladder.py` (3/3); upstream suites per Papers LIII–LVIII.

8 The honest summary

DERIVED: the exact spectrum; mass as closure frequency; the nonrelativistic Coulomb spectrum of hydrogen from the derived equations. CONDITIONAL: the ladder spacing (on the refinement premise P1). *Recorded*: the deviation attributions (nulls, one identity, one observation). CONDITIONAL: the anchor (H-shell-96) and with it m_P and G ; α . PLACED: twelve of thirteen shell integers (postdictions, disclosed). OPEN: the rung-selection rule (fenced: no separable static rule at proof grade, the tested sieves null, the dynamical route behind the couplings), the offset dynamics, the independent hypercharge/mixing, neutrinos. The falsifiable items on record: the pole-mass reformulation should sharpen the table; a future derived sieve must hit all nine shells at density $\leq 1/3$; a derivation of shell 96 that misses measured G kills the anchor.

Reproducibility

Manuscript: `papers/paper-lix/paper-lix.tex`. Sources consolidated: `docs/resonance-mechanism.`
`docs/shell-rule-wo.md`, `docs/sm-closeability-audit.md`; shell table provenance `papers/cascade-`
`§E3`. Run: `python3 scripts/explore_shell_rule.py`; `python3 scripts/explore_geometric_ladder.py`;
`python3 scripts/sm_ledger.py`. Web explorer and its build guide: `web/` in the companion repository.

References

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